Malengo Savings Plan

Product guide and configuration reference. Prepared 15 October 2026.

1. What the plan is

A savings contract. The client pays in whenever they like and takes money out whenever they like, and the balance earns 4% a year, credited at each month end. There is no fixed sum assured: what the client is owed is whatever the account holds. Taking the whole balance out is how the contract ends.

2. The configurable part

Set on the product version when it is published. Changing any of these means publishing a new version; a policy already sold keeps the version it was sold on.

Setting	Malengo	What it does
guaranteedRatePercent	4	Annual rate floor, credited monthly
minimumBalance	0	A withdrawal may empty the account
contributionAllocation%	0	Held back from each deposit
transferAllocation%	0	Held back from a transfer in
monthlyPolicyFee	1,000	Taken at each month end
minEntryAge / maxEntryAge	18 / 60	Enforced when a case is opened
payout amountBasis	ACCOUNT_VALUE	Maturity pays the account, not a face amount
portfolioCode	SAV	An account must be a savings contract

The minimum balance is the one setting that separates Malengo from the Akiba plan beside it: Akiba keeps a 50,000 floor, so its account can never be emptied. Malengo's is zero.

3. Setting it up

1. **Author the product: category ENDOWMENT, portfolio SAV, currency TZS.**
2. **Publish a version carrying the settings in section 2, with a rating table covering ages 18 to** 60 and a MATURITY payout row on ACCOUNT_VALUE.
3. **Open an underwriting case for the client and decide it. Entry age is checked here.**
4. **Issue the policy. The sum assured carries the opening deposit -- see section 5.**
5. **Collect the first premium. That is the opening deposit, and it opens the account.**

Steps 1 and 2 are scripted in backend/scripts/malengo-demo.sh.

4. Running it

A deposit is a premium collection or an ad-hoc top-up; both credit the account in full, since nothing is held back. A withdrawal is asked for by one person and approved by a second, then paid through the payment rail -- and a second withdrawal is refused while one is still in flight. Every movement lands on an append-only ledger: a correction is posted as a reversal, never an edit. At each month end the engine credits interest, then takes the policy fee.

5. Two things to know

- No fixed sum assured, but not a zero one. The wire layer refuses a sum assured of zero, so the contract carries the opening deposit as its face amount. It is a placeholder, not a promise: the maturity payout is ACCOUNT_VALUE. This is the convention the fixed-term deposit already uses.
- Emptying the account does NOT yet end the policy. A withdrawal that takes the balance to zero leaves the policy ACTIVE and the account OPEN, because only the month-end charge run closes an account by exhaustion. Until that is built, a Malengo contract is ended by SURRENDER, which pays the balance and closes the policy properly.

6. What a year looks like

Simulated against a real database through the real engine (MalengoSimulationTest), with every date backdated. A 35-year-old client, 1,000,000 paid in across the year, 150,000 out.

Step	Balance (TZS)
Opening deposit 500,000, then 2 month-ends	**501,284.88**

Top-up 200,000, then 3 more month-ends	**704,418.86**
Withdrew 150,000, then 3 more month-ends	**558,407.20**
Deposited 300,000, then the rest of the year	**866,715.83**
Withdrew the whole balance	**0.00**

850,000 of net cash in became 866,715.83: 4% compounded monthly, less twelve 1,000 fees. After the final withdrawal the policy was still ACTIVE on a zero balance, which is the gap section 5 describes.